

The Alpha-Z Pipeline: Automated Text Normalization for Multi-Generational Filipino Online Content

Arabella N. Bayta
Camarines Sur Polytechnic Colleges
arbayta@my.cspc.edu.ph

Jasmin B. Francisco
Camarines Sur Polytechnic Colleges
jafrancisco@my.cspc.edu.ph

C-Jay B. Lavapie
Camarines Sur Polytechnic Colleges
c-lavapie@my.cspc.edu.ph

Abstract

The rapid growth of social media will keep shaping the way language is spoken in online communication, especially among the young generations in the Philippines. Such informal language as creative spelling, abbreviations, emojis, and code-switching from Filipino to English (Taglish) are becoming increasingly popular in online communication. This evolving linguistic environment presents significant challenges for Natural Language Processing (NLP) systems, which are usually trained on formal text corpus and thus will find it difficult to process non-standard language. Generation Z and Generation Alpha often give rise to new terms of slang, including *korique*, *sizt*, *eme*, *delulu*, and *sksksk*, which are extremely contextual and cultural in their nature. These linguistic differences tend to diminish the precision of downstream NLP applications such as sentiment analysis, topic modeling and content moderation. To overcome this problem, this study presents the Alpha-Z Pipeline, which is an automated text normalization system of multi-generational Filipino web-based texts. The system will work with a hybrid database comprising of curated slang materials in corresponding platforms like Kaggle, GitHub and Hugging Face, and social media text retrieved from Twitter (X), TikTok, Facebook, and Reddit, discussion within Filipino circles. Lexicon-based pre-processing will be combined with sequence-to-sequence models based on transformer, including multilingual T5 and mBERT. Out of vocabulary slang words will be treated using the subword tokenizing method, and the imbalance in the classes will be corrected using data augmentation. The framework proposed will be focused on enhancing NLP performance on informal Filipino social media text and maintaining the semantic meaning and integrity of contexts.

Keywords

Keywords Natural Language Processing, Text Normalization, Filipino Internet Slang, Generation Z Language, Generation Alpha Language, Social Media Text Processing, Transformer Models, Multilingual NLP

ACM Reference Format:

Arabella N. Bayta, Jasmin B. Francisco, and C-Jay B. Lavapie. 2026. The Alpha-Z Pipeline: Automated Text Normalization for Multi-Generational

Filipino Online Content. In . ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

This section introduces the research problem and clearly defines the NLP task being addressed. It explains the significance of the problem, its real-world relevance, and the current limitations of existing approaches.

The fast-paced development of social media platforms has had an enormous influence on how language is employed in online communication. Users are turning to the less formal, dynamic linguistic forms, such as abbreviations, spelling inventions, use of emojis, and code switching between Filipino and English, more often known as Taglish. Such expressions are especially common with younger generations like Gen Z and Gen Alpha, who develop new terms of slang to express humor, identity, and social trends. Such a dynamic and very context-dependent environment poses special problems to natural language processing (NLP) systems, which tend to be trained on standard text corpora and can only cope with non-standard language common to online communities.

Multi-generational differences in slang also increase the problem in the Filipino online environment. As an example, Gen Z users popularized the term '*korique*', '*sizt*', '*eme*', '*nhak*', and the Gen Alpha users have newer expressions, which are influenced by the internet culture, including such terms like '*let him cook*', '*skibidi*', '*delulu*', and '*sksksk*'. Although these lines can be easily interpreted by human readers conversant with the culture of the internet, NLP models frequently fail to represent their intended meanings because of out-of-vocabulary tokens, non-standard spelling, and context-dependent meaning. Therefore, even basic downstream NLP applications like sentiment analysis, topic modeling and content moderation can yield inaccurate results when used on social media data.

The issue of text normalization has become one of the most important preprocessing strategies to overcome these problems. Normalization enables text processing of NLP models more precisely, preserving semantic meaning and enhancing system reliability by replacing non-standard or informal words with standardized equivalents. The current methods of normalization, though, are mostly English-oriented or specialize in English generation to ignore culturally contextual or multilingual situations like the Filipino online communities. This gap limits the applicability of current models in capturing the evolving lexical patterns across multiple generations of social media users.

In order to overcome these limitations, this paper introduces the Alpha-Z Pipeline, an automated text normalization system that will be used to process multi-generational Filipino internet slang. The

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Conference'17, Washington, DC, USA

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

system uses the models of transformers to identify and standardize both Gen Z and Gen Alpha slang expressions without losing grammatical sense and semantic context. Through a combination of lexical and contextual normalization methods, the Alpha-Z Pipeline will offer a powerful solution to process informal Filipino online data to serve a wide variety of applications such as sentiment analysis, conversational agents, and content moderation systems. This paper emphasizes the need to have multi-generational, culturally aware NLP systems that can address the disconnect between the developing online language and computational knowledge.

2 Related Work

This section summarizes previous studies relevant to the research topic, highlighting how others have addressed similar problems, the methods and datasets they used, and the gaps or limitations in their work. It also introduces resources that inform the current study.

A study on the language of various generations suggests the influence of digital space on slang and informal phrases. Sociolinguistic research demonstrates that Filipino Gen Z slang is characterized by innovative morphological processes like clipping, blending, and phonetic play as users adapt the language to online settings. For example, WikaGenZ by Gime and Macascas [5], which records the use of back formation, acronyms, and metathesis by the Filipino Gen Z to construct slang terms indicating sociolinguistic change within the youth of the Metro Manila schools. Similar study by Sawe [8] addresses Filipino slang such as '*Dasurv*' and '*Forda Fer-son*', showing that this slang is not just a form of playful language but also a reflection of digital identity and social conformity among the users of Generation Z. Going outside the Filipino context, comparative studies of Gen Z and Gen Alpha slang have pointed out that they differ in terms of the type of expressions they use, with Gen Alpha being more visually and meme-driven in terms of their slang than Gen Z is in their text-based innovations. These papers define the linguistic diversity of the youth slang but do not discuss the computational processing of this multi-generational changing language, which this text normalization framework of this paper fills.

The study of slang has been a sociolinguistic issue to comprehend how slang is formed and its relevance to a culture. Studies on the semantic and morphological structures of slang show how creativity can influence the use of informal language. The semantic discussion of Gen Z social media slang illustrates the dominance of creative terms, light language formats, abbreviations, and shortenings of social media slang among Generation Z. [9] Research into the morphological structure of Filipino Gen Z slang words reveals that the coinage, borrowing, compounding, blending, and spelling changes processes are underlying mechanisms in how a new slang word gets its formation and is deployed in everyday digital life. The study of slang neologisms is also pursued complementarily by focusing on the dynamic character of slang as a linguistic between-world signifier of cultural identity and immediate experience, in the view that the appearance of slang is a generalized expression of social and cultural shifts. These contributions provide a useful understanding of the processes of slang formation and how it works,

but do not look at the issue of how slang can influence computational processing, and how it can be standardized automatically to be used in NLP, which the Alpha-Z Pipeline attempts to provide. [7]

The informal text processing in social media has been a challenge in natural language processing since most traditional NLP systems are trained on formal texts, like news articles or scholarly texts. Non-standard linguistic phenomena such as misspellings, abbreviations, code switching, and even slang can be a major challenge to computational models. This is indicated in early foundational studies by Baldwin and Li that show that text normalization of social media data yields many positive gains in downstream NLP tasks at part-of-speech tagging and syntactic parsing [2]. These studies made normalization an important preprocessing stage in informal text, and the basis of later computational methods.

Challenges in NLP, such as the findings mentioned by Baldwin and Li, led researchers to examine numerous normalization strategies. Salman Ahmad Ansari et al., have presented optimized nearest-neighbor matching, in which contextual similarity is an important factor in determining which is the most suitable normalized form to use with a non-standard word. [1] Similar study by Bucur et al. casts text normalization as a translation problem that can be solved with multilingual transformer networks like mBART and shows that sequence-to-sequence models can effectively recover both semantic and orthographic expressiveness variation in informal text and downstream NLP problems. The mentioned approaches represent an obvious sequence. [3] The first rule-based or edit-distance models dealt with orthographic variation; transformer-based models take advantage of the contextual and multilingual information by normalizing informal language in a more robust way.

The concept of normalization has been discussed in other studies as applied to integrated NLP systems. Joint text normalization can be conducted in a multi-task learning setting, the normalization can be conducted in parallel with other similar tasks, such as offensive language detection as Shengqiong Wu suggested [10]. Their work has shown that normalization is not simply a preprocessing step but can be integrated into more wide NLP pipelines strategically to improve performance. Emoticons, creative spelling, and slang have been found to negatively influence the sentiment analysis models by Aadil Ganie, and support the practical need of normalization in practical implementation [4]. On the same note, Priyo Agung Prastyo et al. in 2023 established that slang can be normalized so that it enhances the accuracy of sentiment classification, which emphasizes the value of culturally and contextually sensitive normalization methods [6].

In spite of these developments, a majority of existing research concentrates on English or languages with large discursive presence and few studies on code-mixed/culturally specific or Filipino internet slang. The case of Filipino social media is quite different; the language tends to blend Filipino and English (Taglish), contains some regionally specific words, and uses both Gen Z and Gen Alpha multi-generational slang. Current frameworks, either rule-based, sequence-to-sequence, or transformer-based, have not been tailored to address these multilingual, code-switched, and changing linguistic patterns. The proposed study fills this gap by deploying a text normalization framework with an automated framework that is specifically designed to deal with multi-generational Filipino online content. By doing so, it fills both the methodological and linguistic

gaps used by former studies, giving a practical solution towards downstream tasks of sentiment analysis, content moderation, and conversational AI to Filipino online communities.

3 Data and Resources

This section identifies the dataset used in the study and explains its source. It describes the dataset's size, structure, and key features, as well as any preprocessing steps applied.

The proposed study will need a dataset that encompasses the linguistic and informal character of Filipino digital communication. Specifically, the collection should be able to observe slang phrases employed by the members of Generation Z and Generation Alpha that are often used in the social media and often feature creative spelling, shortened forms, and switching between Filipino and English (commonly known as Taglish). To do so, the study will use a combination of the existing slangs datasets with the ones that are gathered recently in the form of text in social media. This will mean that the system will be able to acquire learning of both formal lexical resources and real language that is used in actual online interaction.

The initial part of the dataset is the existing arrays of internet slang and informal language files, which exist in sites like Kaggle, Github, and the Hugging Face Hub. These are often lists of slang terms with their definitions and example sentences of how they are employed in online conversations. Publications of Gen Z and Gen Alpha slang expressions can give ordered correspondences between informal expressions and standardized ones, which are the key to training a normalization system. An example is to have entries of the type mappings of *dasurv* - "deserve," *forda* - "for the," *omsim* - "I swear." These sanctioned materials are the starting point of lexicon of the normalization process and allow the Alpha-Z Pipeline to identify the frequent slang words that are used in Filipino online dialogue.

Although curated lists of slang words are an important source of lexical data, they do not always give examples of usage in full sentences and changing digital communication. In order to overcome this shortcoming, the dataset also receives a boost in terms of custom data retrieval via social media platforms, such as Twitter (X), Tik Tok, Facebook public pages, and discussions on the topic of Filipino communities on Reddit. The use of such platforms is especially topical as the latter are currently popular among younger generations and can be defined as the active environments where new slang terms are originated and popularized. The dataset retrieves the true examples of informal language that includes slang vocabulary, emojis, abbreviations, phonetic spelling, Taglish code-switching, or phonetic spelling by collecting the posts and comments written by the masses on these sites.

The end dataset will consist about more than 20,000 short text samples, each of which will represent a user-generated message in the form of a comment, caption or a tweet. Every piece of data contains the raw informal text and a normalized form of the sentence, which is the standardised form of the text written in standard Filipino, English, or mixed Taglish. Before model training, several preprocessing steps are applied to ensure data quality and consistency. Raw social media text is first cleaned by removing URLs, duplicated posts, and irrelevant metadata while preserving expressive

features such as emojis and repeated characters that may convey emphasis or tone. Tokenization and language identification techniques are then applied to detect Filipino-English code-switching within sentences. Slang terms are identified using the compiled slang lexicon and mapped to their normalized equivalents. The dataset is then structured into input-output pairs, where the input consists of the original informal sentence and the output represents its normalized form. This formulation allows the normalization task to be modeled as a sequence-to-sequence transformation problem, enabling the Alpha-Z Pipeline to learn how to convert informal multi-generational Filipino internet language into standardized text

4 Methodology

The proposed study will be designed as an automated framework to normalize multi-generational Filipino online content, focusing on informal linguistic patterns prevalent among Generation Z and Generation Alpha. These patterns include slang, creative spellings, abbreviations, code-switching between Filipino and English (Taglish), and emoji usage. Such variation often challenges standard NLP models trained on formal corpora, which struggle to handle out-of-vocabulary tokens and non-standard text. To address these challenges, the Alpha-Z Pipeline will combine lexicon-based preprocessing with transformer-based sequence-to-sequence modeling, enabling the system to normalize slang while preserving the semantic meaning of the original text.

The first stage of the pipeline will focus on preprocessing and slang detection. Raw social media posts will be cleaned to remove irrelevant elements such as URLs, duplicate posts, and metadata while retaining expressive markers such as emojis and repeated characters. The text will then be tokenized, and language identification will be applied to separate Filipino, English, or code-mixed segments. Using a curated slang lexicon compiled from existing Generation Z and Generation Alpha datasets, the system will perform initial lexical normalization for terms with known mappings. This lexicon-based step will reduce noise and provide a baseline correction before contextual modeling is applied, allowing the pipeline to recognize commonly documented slang efficiently.

The second stage will employ contextual embeddings derived from transformer models to capture semantic meaning in context. While traditional embeddings such as Word2Vec or GloVe will be considered, the pipeline will primarily utilize transformer-based contextual embeddings from models such as mBERT or multilingual T5, which dynamically represent words depending on their surrounding context. The normalization task will be modeled as a sequence-to-sequence transformation, where the system will "translate" informal social media text into its standardized equivalent. For example, the sentence "soafer delulu si c-jay" will be normalized to "C-Jay is being extremely delusional," preserving meaning while correcting informal expressions.

The pipeline is expected to also address several anticipated challenges. Out-of-vocabulary (OOV) slang terms will be handled using subword tokenization methods such as WordPiece or SentencePiece, which will break unknown words into interpretable units. Class imbalance in slang occurrence will be mitigated through data augmentation and balanced sampling, ensuring that less frequent slang patterns will be represented during training. Additionally,

parameter-efficient fine-tuning of pre-trained transformers will be applied, allowing high performance without excessive computational requirements. By combining lexicon-based preprocessing with transformer-based modeling, the Alpha-Z Pipeline will be able to handle both documented and emerging slang across multiple generations of Filipino internet users.

5 Evaluation Plan

To determine the efficiency of the proposed study, the combination of the quantitative and qualitative metrics will be used to measure the accuracy of the normalization process as well as the maintenance of the semantic meaning. The F1 score as a measure of token-level correctness will be the key measure. Precision will be used to assess the percentage of predicted normalized tokens which are accurate and recall will be used to assess the percentage of real normalized tokens the system will be able to produce. F1 score will be especially suitable since false positives as well as missed corrections may have tremendous impacts downstream with respect to the use of NLP.

Besides the token-level evaluation, the assessment of sequence-level overlap between the normalized output and the reference text will be done by Bilingual Evaluation Understudy (BLEU). BLEU will assess n-gram similarity, and it will permit the research to estimate the similarity of the generated normalized sentences with the anticipated standardized sentences. In a complementary way, ROUGE (Recall-Oriented Understudy for Gisting Evaluation) will also measure the overlap in terms of recalls to make sure that the normalized output does not lose the important meaning of the original sentence. This measure will particularly come in handy in assessment of sentences with code-mixed Filipino-English text and generational expressions.

It will also involve the qualitative analysis to prove the performance of sentences with context-dependent slang or new phrases, especially for Generation Alpha. Normalized outputs will be checked by human evaluators to confirm that the system maintains meaning, readability, and cultural context, which may not be fully reflected by automated measures.

Additionally, it will utilize F1, BLEU, ROUGE, and the qualitative examination of the files, which will thoroughly compare the capability of the Alpha-Z Pipeline to normalise multi-generational and informal online content in the Filipino language. Taken together, these measures will enable the study not only to ensure that the system does not merely replace informal words with standardized equivalents but also to ensure semantic fidelity and fluency, as well as generate text that downstream NLP systems can use in sentiment analysis, topic classification, and content moderation.

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